

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, SPRING SEMESTER 2020-2021

Mathematics for Computer Scientists 2

Time allowed: Two Hours

Answer ALL FIVE Questions

INSTRUCTION FOR STUDENTS:

- The university's honour code applies to this individual short time assessment. The submitted answers **must be your own work** where **answers are attempted without any help from others**. University academic misconduct rules applies here in the case of plagiarism, collusion or false authorship.
- Write the name and student ID at the top of your answer sheets.
- Students are not allowed to ask questions during the short time assessment in accordance with the no shout-out policy.
- You may answer the questions in any order but write the question numbers correctly and clearly.
- At the end of the exam, **upload your** solutions and name the file as **StudentID.pdf** Upload your solution file onto **Moodle**.
- Please ensure that you upload the correct files to the folder labelled as "Final examination submission folder" in Moodle. Request for resubmission will not be accepted after the submission.
- Students should stop writing the answers at the end of exam time. Answer scripts must be submitted to Moodle within **thirty minutes** after the end of exam. In case of any technical issues, the students need to contact the module convener in this period.

IMPORTANT: Late submission will not be accepted.

QUESTION 1: (Relation/Function/Matrices)

- a. Let us consider $x, y \in \mathbb{N}$ and the following relation $x\mathcal{R}y$: $x - y$ is divisible by 4 and $x > y$. Check reflexivity, antisymmetry, and transitivity to assess whether or not this is an order/equivalence relation.

6 Marks

- b. Prove that $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ and then verify it with the following 3x3 matrices.

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}, \text{ and } \mathbf{B} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 2 & 2 & 2 \end{pmatrix}$$

5 Marks

- c. Verify that $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$ with the following 3x3 matrices.

5 Marks

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 5 & 1 \\ 2 & 2 & 0 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 1 & 0 & 3 \\ 1 & -1 & 0 \\ 4 & 2 & 1 \end{pmatrix}$$

- d. Let $\mathbf{B} \in \mathbb{R}_{4 \times 4}$, then identify whether the matrix contains any linear combination of column vectors by means of weighted sum of the other column vectors. Also, find out the scalars.

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & 3 & 7 \\ 2 & 0 & 1 & 5 \\ 1 & 1 & 2 & 5 \\ 0 & 2 & 1 & 3 \end{pmatrix}$$

4 Marks

End of Question 1: Total 20 marks

QUESTION 2: Systems of linear equations Matrix Theory

- a. Check with the help of Cramer's theorem whether the following system of linear equation is a Cramer's System? Also find the solution using cramer's method (if possible).

5 Marks

$$\begin{cases} 2x - y + 2z = 2 \\ x + 2z = 2 \\ 2x - y = 3 \end{cases}$$

- b. Find the ranks of incomplete and complete matrices associated with the following system. On the basis of the calculated ranks assess whether or not the system is compatible.

$$\begin{cases} 2x - y + z = 1 \\ 4x - 2y + 2z = 2 \\ -2x + y - z = 1 \end{cases}$$

5 Marks

- c. Calculate, if possible, the solution or the solutions satisfying the following system of linear equations. If the system allows multiple solutions calculate the number of solutions satisfying the system.

5 Marks

$$\begin{cases} x - y + z = 1 \\ 2x - 2y + 2z = 2 \\ x - z = 0 \end{cases}$$

- d. Find the rank of the complete matrix and calculate, if possible, the solution or the solutions satisfying the following homogenous system of linear equations. If the system allows multiple solutions calculate the number of solutions satisfying the system.

$$\begin{cases} x - y + z = 0 \\ 2x - 2y + 2z = 0 \\ x - z = 0 \end{cases}$$

5 Marks

NOTE: For the above question parts (a, b, c, and d) show and comment on, all the steps in answers.

End of Question 2: Total 20 marks

QUESTION 3: Systems of linear equations Direct Methods

- a. Transform the following system of linear equations into the equivalent triangular system by the method of Gaussian elimination. When the triangular system has been obtained then solve the system (find x, y and z).

6 Marks

$$\begin{cases} x + 2y - 3z = 2 \\ 2x + y - z = 0 \\ x - 4y + 4z = 1 \end{cases}$$

- b. Perform the LU factorisation of the incomplete matrix associated with the following system of linear equations. After the factorisation, solve the original system (find x, y and z) by making use of the LU factorisation.

7 Marks

$$\begin{cases} x + 4y - 3z = 2 \\ 4x + 13y - 14z = 4 \\ -x - 4y + 8z = 8 \end{cases}$$

- c. Consider the following system of linear equations. Apply Gaussian elimination to LU factorise the matrix of coefficients. More specifically, show the equivalence of Gaussian elimination and LU factorisation.

7 Marks

$$\begin{cases} 4x - 3y + 5z = 3 \\ y + 5z = 3 \\ 28x - 20y + 41 = -5 \end{cases}$$

NOTE: For the above question parts (a, b, and c) show and comment on, all the steps in answers.

SECTION B

QUESTION 1: Vector Spaces

- a. Let $(E, +, \cdot)$ be a vector space. Let $(W, +, \cdot)$ and $(S, +, \cdot)$ be two subspaces of $(E, +, \cdot)$. Let $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_u\}$ and $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_v\}$ be two bases of W and S respectively. Then, prove that the sum set $W + S$ is a direct sum $W \oplus S$, if and only if $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_u, \mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_v$ are linearly independent.

14 Marks

Note: To perform the proof, use relevant definitions, theorems, lemmas, corollaries, and propositions studied in this module.

- b. Consider the following vector spaces $(U, +, \cdot)$ and $(V, +, \cdot)$ where

$$U = \{(x, y, z) \in \mathbb{R}^3 \mid \begin{cases} x - y + z = 0 \\ y - 2z = 0 \end{cases}\}$$

$$V = \{(x, y, z) \in \mathbb{R}^3 \mid x - y - z = 0\}$$

- b1. Find the basis of U, V and $U \cap V$ and the dimension of $U + V$.

4 Marks

- b2. What is the geometrical interpretation of $U \cap V$ and $U + V$?

2 Marks

End of Question 1: Total 20 marks**QUESTION 2: Linear Mapping**

- a. Use the additivity and homogeneity properties, and check whether or not the following mappings are linear:

a1. $f(x, y, z) = (2x - y + z, y - z, 3x + z)$ 6 Marks

a2. $f(x, y, z) = (2x + z, y - z, 3 + z)$ 6 Marks

- b. Let f be an endomorphism $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ whose associated matrix A has rank 1. If x, y and z are three linearly independent vectors then check whether or not $f(x), f(y)$ and $f(z)$ are linearly dependent and proportional? Justify your answer.

8 Marks

End of Question 2: Total 20 marks